

Project Report on

Submitted in Big Data Analytics and Business Intelligence Course, CSET 371

**Bachelor of Technology
Of
Bennett University, Greater Noida**

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**BENNETT UNIVERSITY (THE TIMES GROUP), GREATER
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DECLARATION



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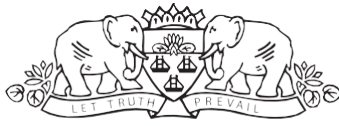
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Chapter 1: Problem Analysis & Introduction

1.1 Background of the Study

Over the past ten years, the food service and restaurant business in India has grown quickly. Multi-outlet chains are becoming more common in cities. It is very hard to analyze how to keep operational efficiency, customer satisfaction, and profitability high across stores that are spread out over a large area. Traditional management systems have a hard time doing this.

Big Data Analytics and Business Intelligence (BI) tools are new technologies that help businesses make sense of large amounts of structured transactional data and turn them into useful, real-time insights. In the restaurant business, this kind of data usually includes daily sales, changes in inventory, customer ratings, service model usage (Dine-In, Delivery, Takeaway), and performance metrics for each outlet.

This project looks at a restaurant chain that has ten locations in five major Indian cities: Bangalore, Hyderabad, Mumbai, Delhi, and Pune. From January 2025 to December 2025, the chain made about INR 12 million in sales. The gross margin was 57.82%, and the cost of food was 42.18%. Even though the overall performance is good, there is a lot of variation in revenue, waste rates, and customer satisfaction ratings between stores. This means that a structured analytical framework is needed.

1.2 Problem Statement

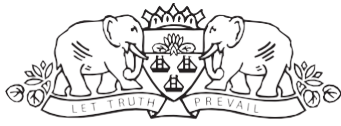
Restaurant chains that operate in more than one city have to deal with the following operational and analytical problems:

- Not having a single, real-time view of revenue, costs, and margins at the outlet level.
- Not being able to tell which menu items make the most money and which ones waste the most.
- There is no systematic way to keep an eye on how demand changes from day to day to make staffing and purchasing more efficient.
- No early-warning systems for inventory waste that causes food costs to go up when they don't have to.
- It's hard to break down stores by how satisfied their customers are so that you can make targeted improvements.

The specific problem addressed by this project is: How can a multi-outlet restaurant chain leverage Power BI-based Big Data dashboards to monitor revenue performance, control food wastage, and identify growth opportunities across cities, service models, and menu items?

1.3 Objectives of the Project

1. To create and build an interactive Power BI dashboard that brings together data from multiple restaurant locations into a single business intelligence platform. To analyse total revenue distribution
2. To look at how total revenue is spread out across cities, outlets, service models, and menu items in



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order to find the main sources of income.

3. To figure out and show gross margin percentages and food cost ratios for both individual stores and chains.
4. To keep an eye on how much inventory is being wasted by store and item, and to send alerts when the amount of waste goes above a certain level.
5. To keep an eye on Month-over-Month (MoM) growth patterns and find times when sales go up because of a holiday or event.
6. To group stores by the average rating of their customers and see how ratings affect sales.
7. To use data-driven insights to make useful business suggestions about pricing, inventory, and outlet strategy.



1.4 Scope of the Project

- Geographic Scope: 10 stores in 5 cities: Bangalore, Hyderabad, Mumbai, Delhi, and Pune.
- Time Frame: An analysis of the whole year, from January 2025 to December 2025.
- The data includes sales transactions, records of opening and closing stock, customer ratings, and service model logs.
- Analytical Scope: Analyzing revenue, gross margin, waste, item performance, day-of-week heatmaps, and month-over-month growth.
- Technological Scope: Microsoft Power BI for making dashboards, DAX for making calculated measures, and Power Query for changing data.

There are some things that the project doesn't cover yet, like predictive machine learning models, real-time data streaming pipelines, and POS system integration. The project does not cover predictive machine learning models, real-time data streaming pipelines, or POS system integration, which are identified as future scope items.

1.5 Review of Existing Systems

There are already a few systems that deal with restaurant analytics, but each one has major flaws compared to the all-in-one BI approach used in this project. Traditional POS Reports show daily sales totals, but they don't let you look at sales across different stores or see how sales are changing over time. Dashboards that use Excel need to have their data combined by hand, and they can't be scaled up. Restaurant ERP systems let you run all of your business from one place, but they are very expensive and don't have a custom BI layer. Google Data Studio lets you make simple charts and graphs, but it doesn't let you track inventory or use many DAX measures. Tableau restaurant templates look great, but they cost a lot to license for small and medium-sized businesses. The suggested Power BI solution is the only one that brings together revenue, margin, rating, and waste analysis on a single platform with automated stock health alerts. None of the other systems we looked at had this feature.



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Chapter 2: Project Design and Methodology

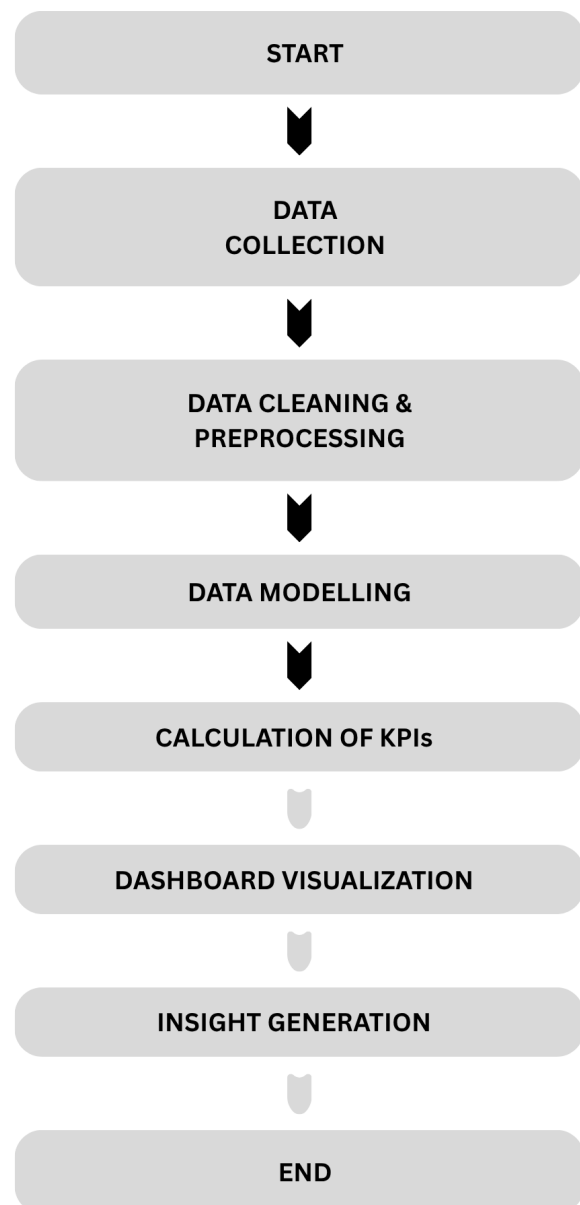
2.1 System Overview

The Restaurant Chain Performance Analysis system is a Business Intelligence architecture with three layers. The Data Layer has raw CSV and Excel files that show daily sales, inventory logs, and customer feedback for all ten locations. The Processing Layer is a Power Query ETL pipeline that cleans, normalizes, and joins datasets. It also has the DAX engine for calculating KPI measures. The Presentation Layer has three interactive Power BI dashboard pages: Executive Overview (KPIs and Revenue), Outlet and Item Analysis, and the Wastage and Inventory Monitor.

The system processes about 365 daily records for each outlet (3,650 total daily rows) from 7 menu items across 10 outlets. This creates a dataset of about 25,550 inventory records for the whole year.

2.2 System Architecture Diagram

The high-level system architecture is based on a standard BI pipeline. Source Systems include logs of point-of-sale transactions, records of managing inventory, and customer reviews. Power Query connectors load flat files into the Power BI data model, which takes care of data ingestion. Data transformation includes handling null values, fixing data types, creating derived columns (Wastage %, Gross Margin %, Rating Band), and making a date table. A Star Schema is used for data modeling. It has a central fact table (transactions) that is connected to dimension tables (Outlet, Item, Date, Service Model). The Analytics Engine uses DAX measures to figure out the Total Revenue, Gross Margin, Food Cost %, MoM Growth %, and Wastage %. Power BI report pages with slicers, bar charts, donut charts, heatmaps, scatter plots, and KPI cards show data in a way that makes it easy to see.



2.3 Proposed Methodology Flow Chart

The project uses the CRISP-DM (Cross-



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Industry Standard Process for Data Mining) method, which has been changed to work with BI. In Phase 1 (Business Understanding), you figure out what the stakeholders want and what the KPIs are. You also make lists of KPIs and wireframes for dashboards. Phase 2 (Data Understanding) looks at datasets and finds problems with their quality. In Phase 3 (Data Preparation), Power Query is used to clean, join, and create derived fields. This makes a cleaned star schema model. In Phase 4 (Modelling), DAX measures and date tables are created. Phase 5 (Evaluation) checks the totals against the source data. In Phase 6 (Deployment), an interactive live dashboard is published to Power BI Service.

2.4 Algorithms Used

The Power BI DAX environment uses the following analytical methods:

SUM, AVERAGE, and COUNT are examples of aggregation functions that work with revenue, margin, and rating KPIs.

For month-over-month and year-over-year growth calculations, use DATEADD and SAMEPERIODLASTYEAR.

Conditional Categorization: Use IF/SWITCH statements to sort Rating Bands into categories (Excellent ≥ 4.5 , Good 4.0–4.49, Average 3.5–3.99, Low < 3.5).

To find the Wastage %, use this formula: $(\text{Opening Stock} - \text{Closing Stock}) / \text{Opening Stock} * 100$. The percentage of food cost is the total food cost divided by the total revenue times 100. The gross margin percentage is the revenue minus the food cost divided by the revenue times 100.

Ranking: The RANKX function lets you rank outlet and item revenue in visuals.

Heatmap Encoding: Using color gradient scales to apply conditional formatting to the day-of-week revenue matrix.

2.5 Tools and Technologies Used

Microsoft Power BI Desktop is the main tool for making, visualizing, and publishing dashboards.

Power Query (M Language): A pipeline for getting data out, changing it, and loading it.

DAX, or Data Analysis Expressions, is a way to use the Vertipaq in-memory engine to measure KPIs and create calculated columns.

Microsoft Excel / CSV: Files that contain raw data about transactions and inventory.

Power BI Service is a cloud-based platform for sharing and publishing dashboards.

Star Schema Data Model: A dimensional design pattern that makes analytical queries run faster.



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2.6 Experimental Setup / Environment

The development environment includes Windows 10/11, Microsoft Power BI Desktop (the most recent version is from 2025), local CSV/Excel files that are imported through Power Query, the built-in Vertipaq in-memory columnar engine for DAX, the Power BI visual layer with custom conditional formatting, and Power BI Service (app.powerbi.com) for publishing.



Date	Outlet	City	Model	Item	Category	Quantity	Unit_Price	Revenue	Food_Cost	Gross_Ma	Zomato_R
2025-09-1	Andheri	Mumbai	Delivery	Burger	Fast Food	33	206	6798	3202.2045	3595.7954	3.7
2025-05-2	Andheri	Mumbai	Takeaway	Sandwich	Fast Food	115	170	19550	6868.5650	12681.434	4.7
2025-11-2	Banjara_H	Hyderabad	Delivery	Pasta	Italian	105	288	30240	10808.075	19431.924	4.6
2025-03-0	Andheri	Mumbai	Dine-In	Fries	Fast Food	94	114	10716	4827.3043	5888.6956	4.5
2025-04-1	Gachibow	Hyderabad	Delivery	Burger	Fast Food	38	218	8284	3043.4571	5240.5428	3.9
2025-09-0	Saket	Delhi	Dine-In	Sandwich	Fast Food	45	189	8505	3264.5068	5240.4931	4.6
2025-02-1	Whitefield	Bangalore	Delivery	Burger	Fast Food	27	181	4887	2333.9424	2553.0575	4.1
2025-06-2	Hinjewadi	Pune	Delivery	Pasta	Italian	28	309	8652	4130.1298	4521.8701	4.4
2025-09-0	Hinjewadi	Pune	Dine-In	Fries	Fast Food	20	107	2140	1155.2929	984.70709	3.5
2025-09-0	Gachibow	Hyderabad	Takeaway	Salad	Healthy	114	188	21432	11162.536	10269.463	4.4
2025-04-2	Banjara_H	Hyderabad	Takeaway	Fries	Fast Food	32	129	4128	2169.1517	1958.8482	3.5
2025-10-0	Andheri	Mumbai	Delivery	Burger	Fast Food	41	224	9184	3619.3336	5564.6663	3.6
2025-12-2	Andheri	Mumbai	Takeaway	Burger	Fast Food	115	225	25875	13751.486	12123.513	4.8
2025-12-0	Saket	Delhi	Dine-In	Sandwich	Fast Food	49	148	7252	3802.2681	3449.7318	4.3
2025-11-2	Indiranaga	Bangalore	Dine-In	Pasta	Italian	104	295	30680	13836.895	16843.104	4.5
2025-01-1	Connaugh	Delhi	Dine-In	Pizza	Italian	44	309	13596	7391.9775	6204.0224	4.3
2025-05-1	Koregaon	Pune	Delivery	Salad	Healthy	64	161	10304	4782.7505	5521.2494	3.5
2025-09-2	Andheri	Mumbai	Takeaway	Burger	Fast Food	22	198	4356	1634.3510	2721.6489	4.2
2025-08-1	Banjara_H	Hyderabad	Dine-In	Fries	Fast Food	67	131	8777	4257.7012	4519.2987	4
2025-12-0	Gachibow	Hyderabad	Delivery	Salad	Healthy	39	172	6708	2802.6304	3905.3695	3.9
2025-10-1	Saket	Delhi	Delivery	Pizza	Italian	109	308	33572	12896.966	20675.033	4.6
2025-01-0	Andheri	Mumbai	Delivery	Burger	Fast Food	70	201	14070	6173.9358	7896.0641	3.9
2025-07-1	Whitefield	Bangalore	Delivery	Pasta	Italian	99	300	29700	14382.914	15317.085	3.8
2025-07-1	Bandra	Mumbai	Dine-In	Fries	Fast Food	100	146	14600	6928.7876	7671.2123	3.9
2025-05-1	Indiranaga	Bangalore	Delivery	Pasta	Italian	60	303	18180	7807.9377	10372.062	4.6

2.7 Dataset Description

The project makes use of two main sets of data. Dataset 1 (Transaction and Revenue Data) has the following fields: Date (the date of the transaction), Outlet (the name and location of the outlet), Item (the menu item), Service_Model (Dine-In/Delivery/Takeaway), Revenue (the daily revenue in INR),



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Food_Cost (the daily food cost in INR), and Customer_Rating (the average daily rating on a 1–5 scale). There are five fields in Dataset 2 (Inventory and Wastage Data): Outlet, Item,

Date	Outlet	Item	Opening_Stock	Received_kg	Sold_kg	Wastage_kg	Closing_Stock_kg
2025-05-21	Bandra	Burger	111	108	85	3	131
2025-10-12	Bandra	Pizza	126	63	129	6	54
2025-06-18	Gachibowli	Fries	144	50	162	6	26
2025-03-26	Koregaon_Park	Fries	135	91	126	2	98
2025-11-06	Andheri	Wrap	155	55	73	2	135
2025-08-05	Indiranagar	Burger	123	92	92	4	119
2025-02-23	Gachibowli	Burger	188	100	125	5	158
2025-04-25	Indiranagar	Fries	179	56	117	3	115
2025-09-19	Whitefield	Pasta	88	62	151	5	-6
2025-02-15	Andheri	Sandwich	137	41	77	4	97
2025-10-20	Connaught_Place	Burger	87	75	89	5	68
2025-04-25	Connaught_Place	Pizza	188	83	129	3	139
2025-01-05	Andheri	Salad	97	45	108	2	32
2025-01-27	Andheri	Burger	171	100	145	5	121
2025-11-09	Indiranagar	Pasta	185	47	106	3	123
2025-11-20	Koregaon_Park	Pasta	160	108	121	4	143
2025-03-13	Andheri	Wrap	127	63	101	3	86
2025-01-04	Bandra	Fries	106	90	70	2	124
2025-11-06	Saket	Pizza	193	40	138	5	90
2025-01-05	Andheri	Fries	129	119	158	7	83
2025-02-07	Indiranagar	Wrap	99	102	166	4	31
2025-04-05	Koregaon_Park	Salad	154	89	150	11	82
2025-12-14	Koregaon_Park	Pizza	161	42	131	7	65
2025-11-23	Bandra	Sandwich	128	107	124	3	108
2025-03-14	Andheri	Wrap	86	109	118	4	73

Opening_Stock_kg, Closing_Stock_kg, and Wastage_% (a derived field). The total opening stock for the chain was 2,82,688 kg, and the total closing stock was 2,76,740 kg. This means that the chain lost about 5,948 kg, or 6.97% of its stock.

Sample data for Sales Table:

Sample data for Inventory Table:

Chapter 3: Project Implementation

3.1 Development Environment

Power BI desktop, version 2025, was used to create this project on Windows 10/11. Local files with .csv or .xlsx extensions were loaded into the platform as sources of information via Power Query connector. The analytical part is executed with the help of Vertipaq in-built technology that executes DAX measures upon a certain query request. Those who were expected to view the final dashboards were able to do it through Power BI service.

3.2 Module Description

Module 1 - Executive Overview Dashboard: Consists of five key performance indicators: Total Revenue: 12M, Wastage %: 6.97%, Food Cost %: 42.18%, Gross Margin %: 57.82%, and Gross Margin Value: 6.71M. It contains a bar graph of Revenue by City, donut graph of Revenue by Service Model, time series graph of Revenue by Date (January - December 2025), bar graph of Revenue by Outlet, and MoM Growth % graph.

Module 2 - Outlet & Item Performance Dashboard: This module has an outlet-wise summary table containing Average Rating, Gross Margin, Food Cost %, and Total Revenue for all 10 outlets. It has a revenue heatmap by day of the week with colour conditionality (from green to red). It has a bubble graph (Revenue vs Gross Margin by Item), bar graph of Total Revenue by Item, and bar graph of Revenue by Rating Band.

Module 3 - Wastage & Inventory Monitor: This dashboard has a wastage detail table of outlets and items with colour coding of Wastage % (green <7%, amber 7-10%, red >10%). This module has a bar graph of Total Wastage by Outlet, Stock Health alert card, and Target Wastage Cost key performance indicator (target = 1M, actual = 24K).



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3.3 Code Implementation (Key DAX Measures)

Total Revenue

Total Revenue = SUM(Transactions[Revenue])

Gross Margin %

Gross Margin % = DIVIDE([Total Revenue] - [Total Food Cost], [Total Revenue]) * 100

Wastage %

Wastage % = DIVIDE(SUM(Inventory[Opening_Stock_kg]) - SUM(Inventory[Closing_Stock_kg]), SUM(Inventory[Opening_Stock_kg])) * 100

MoM Growth %

MoM Growth % = DIVIDE([Total Revenue] - CALCULATE([Total Revenue], DATEADD('Date'[Date], -1, MONTH)), CALCULATE([Total Revenue], DATEADD('Date'[Date], -1, MONTH))) * 100

Rating Band

Rating Band = SWITCH(TRUE(), [Avg Rating] >= 4.5, "1 - Excellent", [Avg Rating] >= 4.0, "2 - Good", [Avg Rating] >= 3.5, "3 - Average", "4 - Low")

3.4 System Interface / Screenshots

The dashboard has been created using Microsoft Power BI Desktop and consists of three pages as mentioned in Section 3.2. The following screenshots illustrate the three dashboard components that have been created for this task.

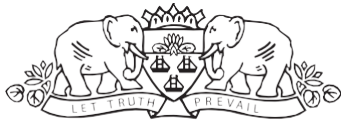
Figure 3.1 – Executive Overview Dashboard

The Executive Overview Dashboard give an overview of overall restaurant performance across all outlets. It helps the concerned decision-makers to quickly analyse important financial and operational metrics.

Key Performance Indicators (KPIs)



Total Revenue (12M): Calculated by sum of revenue represents the overall revenue generated by all outlets



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Wastage Percentage (6.97%): Represents the amount of total raw materials wasted, highlighting operational inefficiency.

Food Cost Percentage (42.18%): Shows the percentage of revenue spent on raw materials for the outlets.

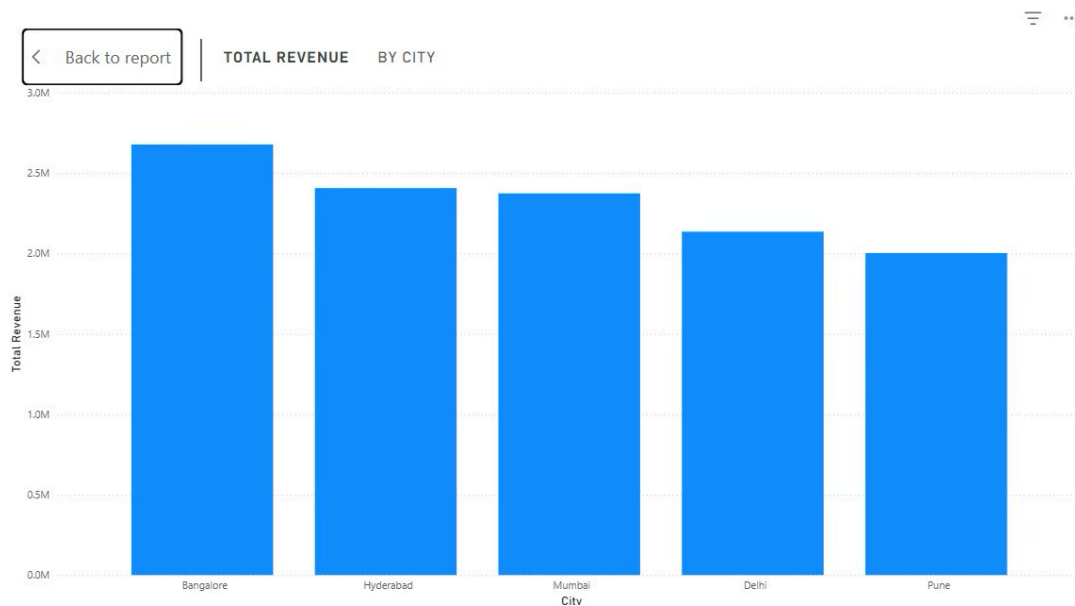
Gross Margin Percentage (57.82%): Represents percentage of profitability after deducting the cost of raw materials from revenue.

Gross Margin (6.71M): Displays the profit generated from the revenue after deducting the cost of raw materials.

Visual Insights

Total Revenue by City:

This bar chart represents the revenue across different cities. It helps identify high-performing regions (e.g., Bangalore) and low-performing regions (e.g., Pune).

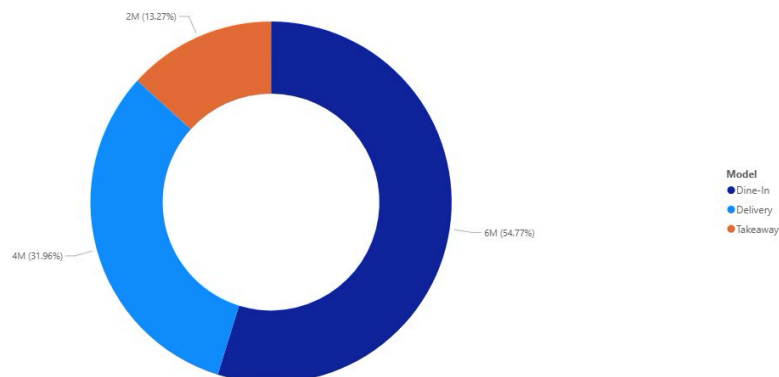


Revenue by Business Model:

This donut chart represents the share of revenue across Dine-in, Delivery, and Takeaway modes. It highlights the high contribution of Dine-in services (54.77%).

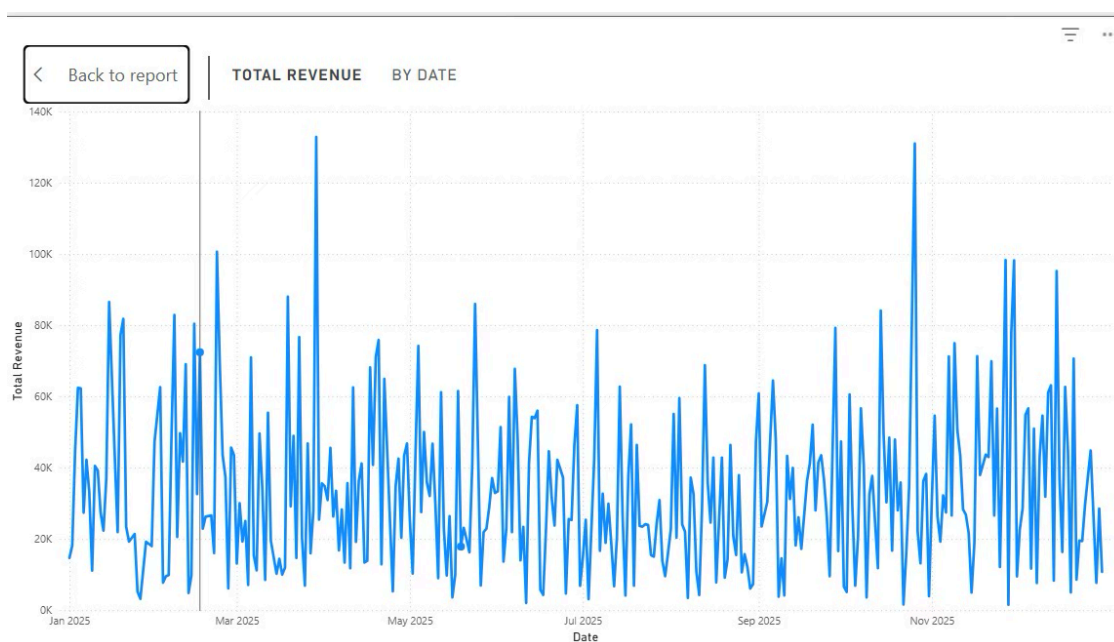
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SUM OF REVENUE BY MODEL



Revenue Trend Over Time:

This line chart shows daily revenue fluctuations by date, helping identify seasonal patterns, peak periods, and demand variability.



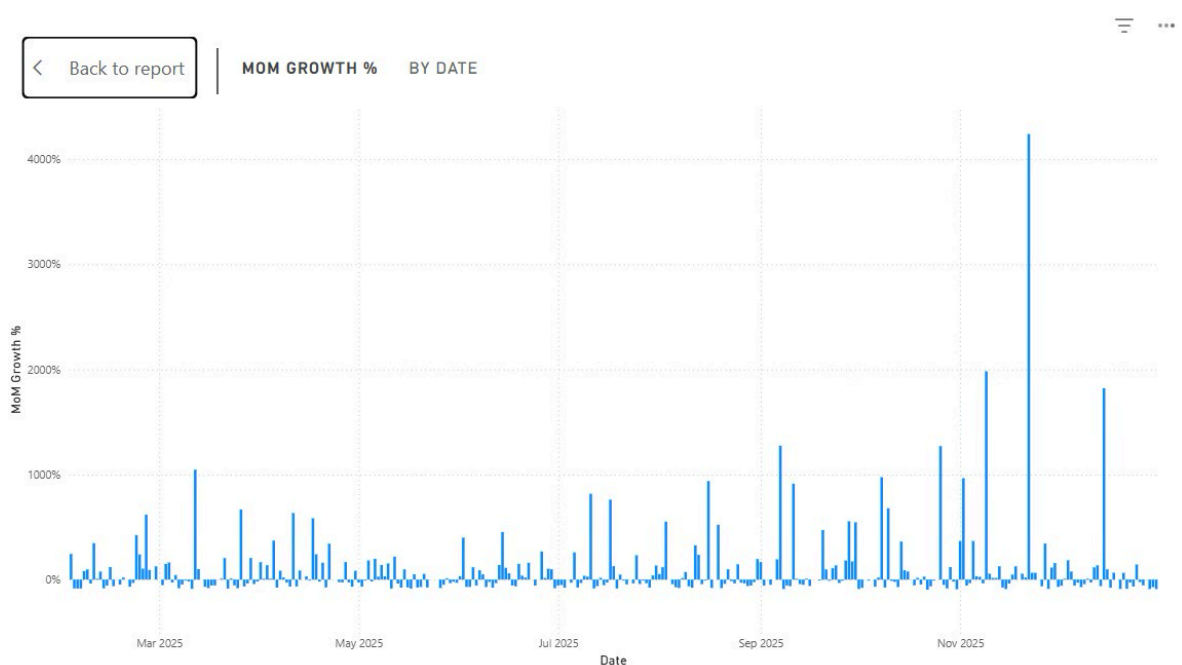
Top Revenue by Outlet:

This bar chart represents a comparison at an outlet-level performance, enabling identification of top-performing (Indiranagar) and underperforming (Koregaon Park) outlets.



Month-on-Month Growth (%):

This line chart represents growth trends over time. Sudden spikes or drops indicate instability and require further analysis.





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Figure 3.2 – Outlet and Item Performance Dashboard

This page focuses on driving insights at the outlet and menu level to draw conclusion regarding key revenue generators and profitability contributors.

Visual Insights

Outlet Performance Table:

This table includes Average Rating, Gross Margin, Food Cost Percentage, Total Revenue.

It allows the comparison of outlets based on the financial performance and customer satisfaction.

Outlet	Avg Rating	Sum of Gross_Margin	Food Cost %	Gross Margin %	Total Revenue
Andheri	4.19	7,42,778.15	41.97%	58.03%	1279954
Bandra	4.10	6,36,863.60	41.80%	58.20%	1094195
Banjara_Hills	4.05	6,27,474.31	42.14%	57.86%	1084466
Connaught_Place	3.89	5,67,815.22	42.08%	57.92%	980276
Gachibowli	4.20	7,60,369.69	42.51%	57.49%	1322572
Hinjewadi	4.06	6,57,624.14	42.64%	57.36%	1146554
Indiranagar	4.29	7,99,677.43	42.57%	57.43%	1392529
Koregaon_Park	3.89	4,96,127.30	42.07%	57.93%	856394
Saket	4.15	6,73,934.50	41.69%	58.31%	1155801
Whitefield	4.16	7,43,908.20	42.14%	57.86%	1285697
Total	4.10	67,06,572.54	42.18%	57.82%	11598438

Day-wise Revenue Heatmap:

The heatmap shows the revenue distribution across days of the week for each outlet. It highlightst the following:

- Busiest business days (e.g., weekends)
- Variations in outlet performance

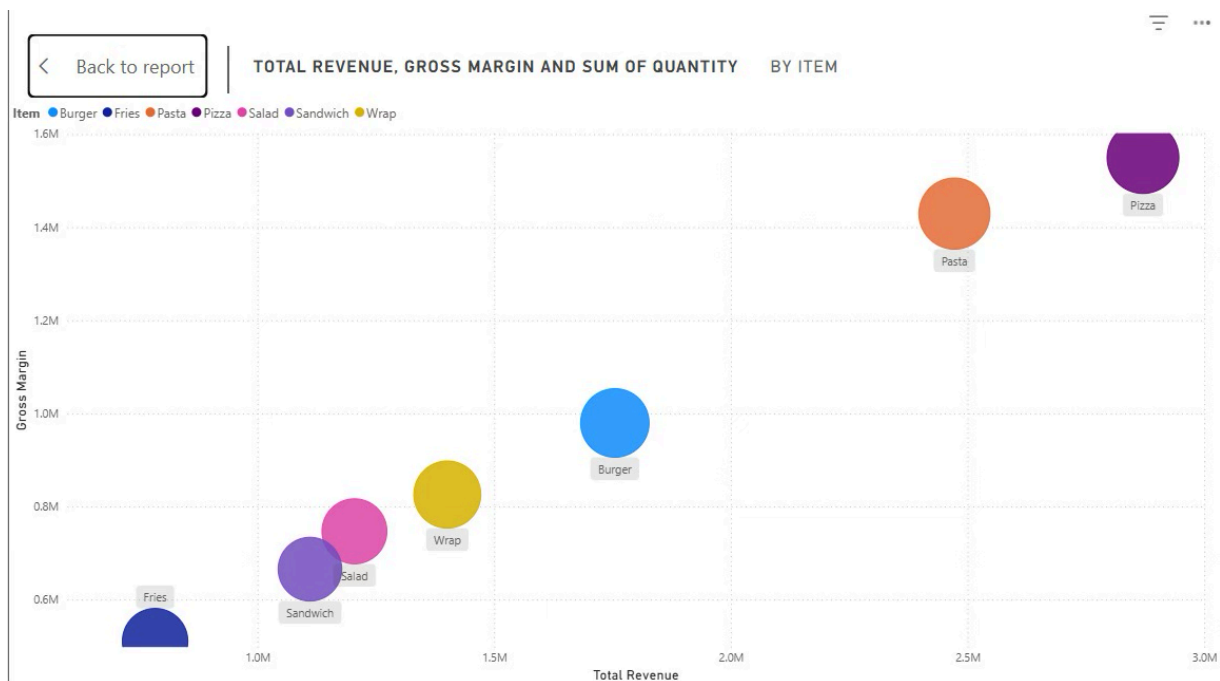
Item-Level Profitability Analysis:

Bubble Plot (Total Revenue vs Gross Margin):

This visualization analyses the profitability of menu items:

Outlet	Fri	Mon	Sat	Sun	Thu	Tue	Wed	Total
Andheri	126467	159294	124696	402398	129971	218328	118800	1279954
Bandra	127324	170130	111114	182606	195939	135750	171332	1094195
Banjara_Hills	221818	56165	216070	163174	169208	130011	128020	1084466
Connaught_Place	196154	150692	100503	201194	82237	174687	74809	980276
Gachibowli	205321	157068	297124	208515	168472	160169	125903	1322572
Hinjewadi	135633	91084	231815	254387	180576	108220	144839	1146554
Indiranagar	238673	214729	230022	205332	174803	187989	140981	1392529
Koregaon_Park	159201	120985	136236	197062	109722	49313	83875	856394
Saket	179127	171008	230983	136160	175817	112979	149727	1155801
Whitefield	151540	170828	189150	281709	155136	223651	113683	1285697
Total	1741258	1461983	1867713	2232537	1541881	1501097	1251969	11598438

- High Revenue with High Margin → Top-performing items
- High Revenue with Low Margin → Improvement gap
- Low Revenue with Low Margin → Can be removed



Revenue by Item:

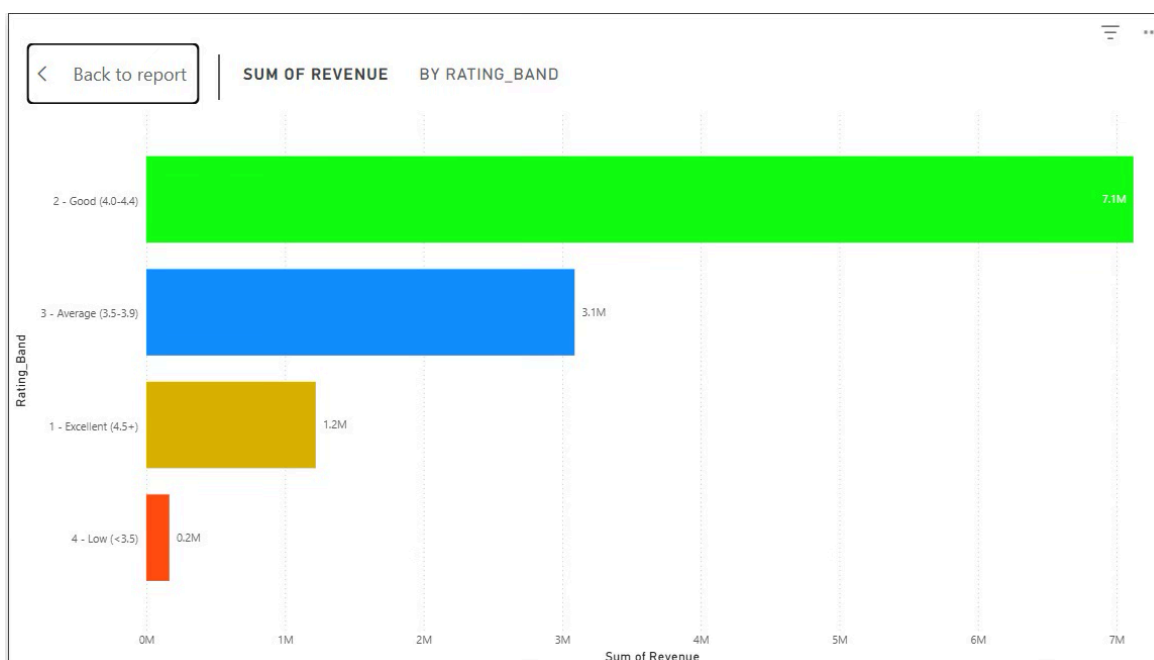
This bar chart shows contribution of each item to the total revenue. Items such as Pizza and Pasta contribute significantly, whereas fries contribute the least.



Revenue by Rating Band:

This chart categorizes the total revenue based on customer ratings (Excellent, Good, Average, Low).

It helps analyze the relationship between customer satisfaction and revenue generation.



Outlet	Item	Sum of Opening_Stock_kg	Sum of Closing_Stock_kg	Wastage %
Connaught_Place	Salad	4,580.00	4,484.00	14.51%
Koregaon_Park	Sandwich	6,227.00	6,150.00	13.76%
Koregaon_Park	Pizza	5,190.00	5,080.00	13.70%
Connaught_Place	Pizza	5,293.00	5,197.00	13.20%
Koregaon_Park	Burger	4,949.00	4,811.00	13.10%
Koregaon_Park	Salad	4,079.00	4,087.00	12.86%
Connaught_Place	Burger	4,810.00	4,683.00	11.52%
Bandra	Salad	4,315.00	4,305.00	10.87%
Koregaon_Park	Pasta	5,240.00	5,158.00	10.69%
Connaught_Place	Fries	4,855.00	4,863.00	9.73%
Koregaon_Park	Wrap	4,770.00	4,715.00	9.68%
Koregaon_Park	Fries	4,790.00	4,694.00	9.60%
Connaught_Place	Wrap	4,919.00	4,931.00	9.35%
Connaught_Place	Sandwich	5,018.00	4,903.00	9.35%
Banjara_Hills	Salad	3,833.00	3,782.00	9.32%
Hinjewadi	Salad	3,739.00	3,727.00	9.18%
Banjara_Hills	Pizza	4,449.00	4,323.00	9.11%
Andheri	Salad	3,831.00	3,711.00	9.11%
Bandra	Pizza	4,467.00	4,343.00	9.03%
Saket	Salad	3,957.00	3,839.00	8.96%
Whitefield	Salad	3,737.00	3,597.00	8.74%
Banjara_Hills	Fries	5,199.00	5,097.00	8.64%
Banjara_Hills	Burger	4,304.00	4,324.00	8.62%
Connaught_Place	Pasta	4,623.00	4,533.00	8.31%
Bandra	Burger	4,274.00	4,129.00	7.89%
Hinjewadi	Pizza	3,918.00	3,912.00	7.79%
Hinjewadi	Burger	3,935.00	3,823.00	7.22%
Total		2,82,688.00	2,76,740.00	6.97%

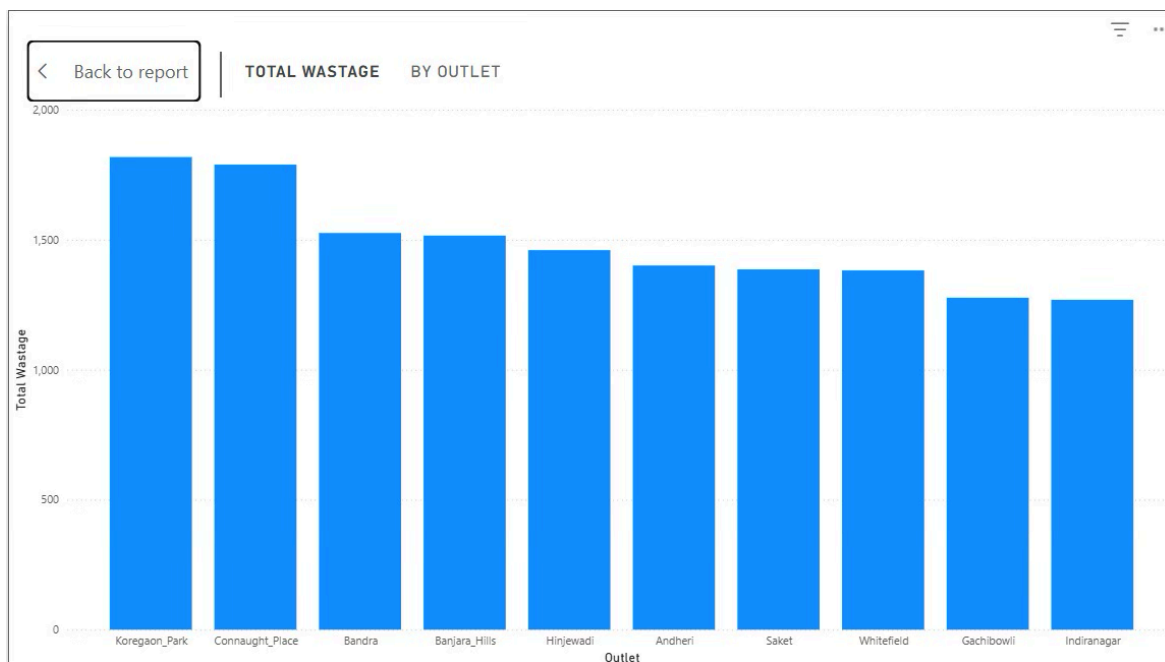
Figure 3.3 – Wastage and Inventory Monitor Dashboard

This dashboard page is made to monitor inventory efficiency, derive insights about wastage and control cost leakage due to wastage.

Visual Insights

Inventory Tracking Table:

This table displays the Opening Stock, Closing Stock, and Wastage Percentage





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This helps in tracking of material consumption and identification of inefficiencies at item and outlet levels.



Wastage by Outlet:

This bar chart compares wastage across various outlets.

Outlets with a higher wastage percentage need operational guidance and improved inventory management practices.

Stock Monitoring and Alerts:

This dashboard includes alert indicators such as:

- Stock health status: It represents the overall condition of inventory across all outlets.
- Number of outlets with inventory issues: It shows the total number of outlets currently facing inventory related problems.

These alerts help management quickly identify critical operational risks.

Wastage Cost vs Target:

This section compares actual wastage cost with the target value.

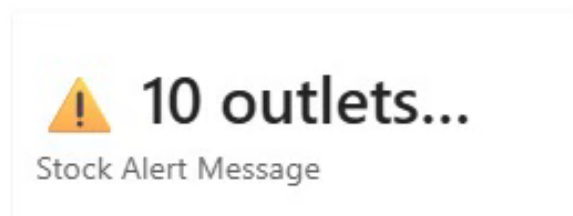
A significant deviation indicates inefficiency and highlights the need for corrective actions.

Target Wastage Cost:

It is the benchmark value for acceptable wastage. It acts as a reference for performance evaluation.



Wastage_Cost_Rs and Target
Wastage Cost by Date



1M
Target Wastage Cost



Chapter 4: Project Results and Discussion

4.1 Experimental Setup

The test environment comprises of the full year 2025 data set for the restaurant chain. It is the restaurant chain's data model that links up transaction facts to dimensions such as Outlet, Item, Date, and Service Model. All the five KPI measures have been tested using the total figures on Power BI compared to their equivalents in Excel.

4.2 Output Results

Overall KPIs: Total Revenue = INR 12,000,000 (12M); Gross Margin % = 57.82%; Food Cost % = 42.18%; Wastage % = 6.97%; Gross Margin Value = INR 6,710,000 (6.71M).

Total Revenue by City: Bangalore at approx. INR 2.7M; Hyderabad at 2.5M; Mumbai at 2.5M; Delhi at 2.3M; Pune at 2.1M.

Revenue by Service Mode: Dine-in service is 54.77% (approx. INR 6.57M); Delivery is 31.96% (approx. INR 3.84M); Takeaway mode contributes approx. INR 1.59M.

Top Performing Outlet: By revenue and rating: Indiranagar – approx. INR 13,92,529, Rating: 4.29; Gachibowli – approx. INR 13,22,572, Rating: 4.20; Whitefield – approx. INR 12,85,697, Rating: 4.16; Andheri – approx. INR 12,79,954, Rating: 4.19; Saket – approx. INR 11,55,801, Rating: 4.15.

Revenue by Menu Item: Pizza – approx. INR 2.9M

4.3 Performance Analysis

Waste analysis: Critical wastages above 10 percent belong to Connaught Place Salad (14.51%), Connaught Place Pizza (13.20%), Connaught Place Burger (11.52%), and Bandra Salad (10.87%). There is an issue with systematic wastages in Connaught Place that need to be urgently resolved via better inventory management. An overall chain wastage of 6.97% is directly proportional to the 42.18% food cost ratio.

MoM growth analysis: Revenues have been fairly steady in terms of month-to-month patterns from March to October 2025; however, there is a sharp spike in November 2025 in month-to-month growth rates above 4000%. This needs to be investigated for reasons such as a large promotional drive or festive season sales.

Margin analysis: The company has maintained consistent gross margins between 57.36% and 58.31% across all the stores, where Hinjewadi is at the lower end, and Saket is at the higher end. A very thin margin difference of only 0.95% across all 10 stores points towards a standard food cost ratio and implies that the revenues are what will drive profitability.

4.4 Discussion

- **Revenue Contribution:** The top 3 branches (Indiranagar, Gachibowli, Whitefield) provide a 34% revenue contribution from only 30% of the branches, which means that good locations with high-quality customer experience can add to revenue.
- **Rating & Revenue:** It has been observed that higher ratings (4.15+) always lead to revenues above average, which again validates the importance of investing in customer experience.
- **Profitability through Wastage Management:** Wastage, which is currently 6.97%, will help in improving the gross profit margin of the restaurant when reduced by even a mere 1%.
- **Leveraging Service Model:** Delivery model has contributed only 31.96%, which indicates its huge potential to boost overall revenue through better management during low demand days such as Wednesday.
- **Item Revenue Strategy:** Pizza & Pasta items have provided approximately 47% of the total revenue through items sold. Items like Pizza & Pasta should be protected and other wasteful items, like Salad in Connaught Place, must be minimized.

Chapter 5: Project Conclusion and Presentation

5.1 Conclusion

The Big Data Analytics and Business Intelligence dashboard has been successfully created in this project to analyze data for a chain of 10 restaurants using Microsoft Power BI. The dashboard analyzes data on revenue, profit margins, customer ratings, and wastage of inventory for the purpose of data-based decisions.

Conclusions from the data include: Bangalore being the best city in terms of revenue generation; Indiranagar having the best performance in terms of both revenue generation (INR 13.9 lakhs) and customer rating (4.29 out of 5); the Dine-in type being the most popular (54.77%); the Pizza & Pasta items generating about 47% revenue from sales; and Connaught Place having maximum wastage risk, especially in case of the salad (14.51%).

5.2 Limitations

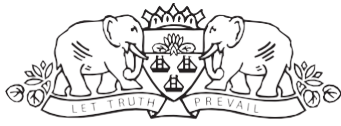
- The dashboard has been constructed based on historical data and thus does not support real-time analytics.
- The inclusion of predictive demand forecasts and recommendations based on machine learning has not been considered yet.
- The demographic data such as customer age and gender was unavailable in the database.
- The sudden increase in Month Over Month growth in November 2025 needs to be explored further.
- The POS system integration has not been done yet.

5.3 Future Scope

- Integration with real-time POS systems by implementing Power BI streaming datasets to provide instant data refresh for dashboards.
- Implementation of ML-driven demand forecasting models (ARIMA and Prophet) for weekly demand predictions to determine order quantities.
- Customer segmentation using RFM analysis (Recency, Frequency, Monetary value) based on loyalty programme information.
- Automated wastage alert emails sent through Power Automate when the wastage of individual items surpasses threshold levels.
- Incorporation of competitor performance comparison data as well as macroeconomic variables (such as inflation rates and cost of transportation).
- Provision of mobile Power BI app for accessing KPIs from smartphones.

5.4 References

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4. National Restaurant Association India (NRAI). (2024). India Food Services Report. NRAI Publications.
5. Han, J., Pei, J., & Tong, H. (2022). Data Mining: Concepts and Techniques (4th ed.). Morgan Kaufmann.
6. CRISP-DM Consortium. (2000). CRISP-DM 1.0: Step-by-step data mining guide. SPSS Inc.



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Appendix A – Additional Screenshots



The succeeding pages include the three Power BI dashboard screenshots relevant to the aforementioned modules presented in Chapter 3. Please provide the dashboard screenshots for each module where indicated below.

Figure A.1 – Executive Overview Dashboard (Page 1)

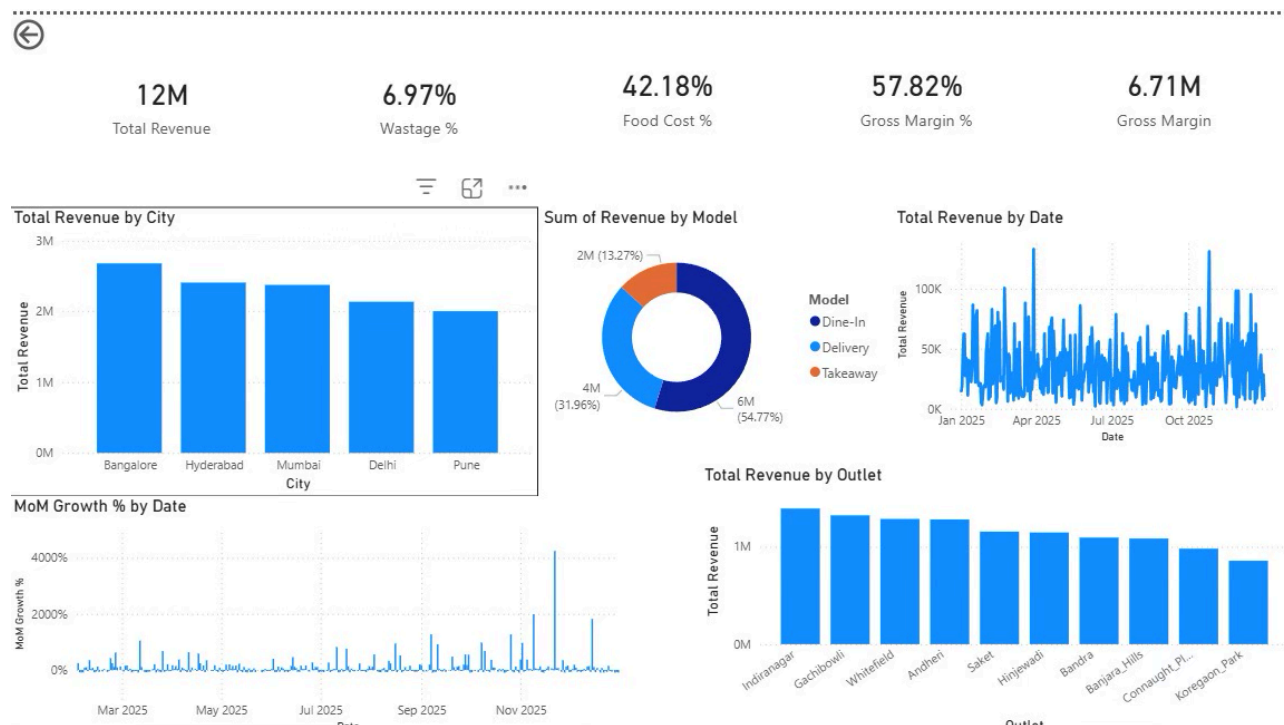


Figure A.2 – Outlet and Item Performance Dashboard (Page 2)



Figure A.3 – Wastage and Inventory Monitor Dashboard (Page 3)



